TrustSphere AI — Digital Evidence Verification Report

Generated 2026-10-23 16:43:03 UTC · Case ID 1bf1a79f-e5de-4b54-a3a9-ee697c444bd3

1. Evidence Identification

Original filename	2_receipt_TAMPERED.png
File type	image
SHA-256 hash	`312161ce20aaa7f2f4bdafeaf1a134e0a4e9be764a472241fbac65c682ba4f52`
Uploaded (UTC)	2026-10-23T16:43:03.850001
Analysis status	done

2. Forensic Verdict

INCONCLUSIVE — overall authenticity score 0.51 (0 = fake, 1 = authentic)

Manipulation type	Spliced / Copy-Move Forgery (confidence 0.81)

Reasoning:
• Overall authenticity score 0.51 → Inconclusive.
• Error-level analysis shows inconsistent compression regions (splice).

3. Agent Findings

metadata — authenticity 0.450

Check	Status	Note
camera_info	suspicious	No camera make/model — common in AI-generated or re-encoded images
software	ok	No editing software tag
timestamps	suspicious	No EXIF timestamps present

image_forensics — authenticity 0.550

Check	Status	Note
ela	suspicious	Mean error level 24.33 (heatmap saved)
fft	ok	High-frequency energy ratio 0.282
deep_model	ok	Deepfake classifier authenticity 0.505
xai_gradcam	ok	Model-attention heatmap saved (gradcam)

security — authenticity n/a (no signal)

Check	Status	Note
c2pa_manifest	ok	No C2PA content credentials embedded (typical for most media).
invisible_watermark	skipped	Image too small for DWT-DCT decode.

4. Explainability Exhibits (XAI)

Exhibit — metadata feature importance (exact Shapley contributions; green → authentic, red → fake)

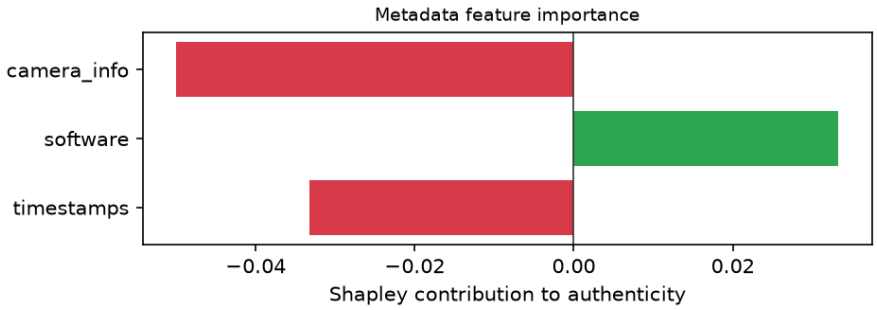


Exhibit — image_forensics: ELA heatmap (bright regions compress inconsistently → possible splice)

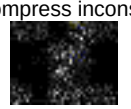
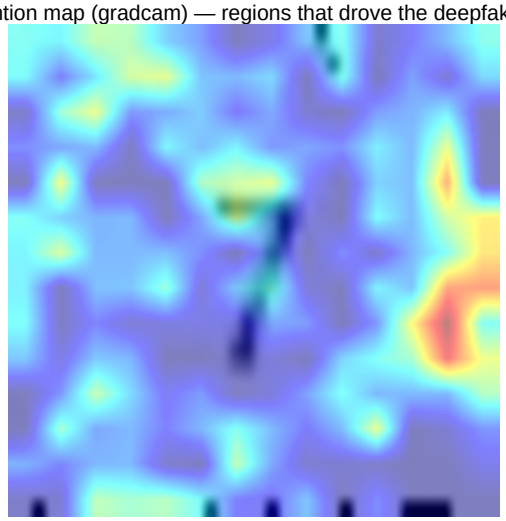


Exhibit — image_forensics: model attention map (gradcam) — regions that drove the deepfake classifier's decision



5. Chain of Custody

Event	Timestamp (UTC)	Detail
uploaded	2026-10-23T16:43:03.850001	
analyzed	2026-10-23T16:43:03.951294+00:00	
signed	2026-10-23T16:43:03.951437+00:00	`6217864d3e24af8f2157b4800b3b29c95f497c141297c36aae97212c7b77f184`
blockchain_skipped	2026-10-23T16:43:03.951452+00:00	`POLYGON_PRIVATE_KEY not set — anchoring disabled`

Blockchain network	polygon-amoy
Anchor status	skipped

6. Record Integrity — Digital Signature

The analysis record below is signed with the TrustSphere server's Ed25519 key. Any alteration of the signed payload invalidates the signature (verifiable via GET /api/verify/{file_id}).

Algorithm	Ed25519
Key fingerprint	`6217864d3e24af8f2157b4800b3b29c95f497c141297c36aae97212c7b77f184`
Signature	`aabb750047860e7ac8507aa1e42932b9564acc833ed86190b7b130bd2e331f3462d54c83174ec9eb74abbe380cdae7823a276f8eccdcc5c503ee62661f1ced09`